HACKATHON – FEST 3.0

- **Problem Statement ID –** Healthcare PS1
- **Problem Statement Title-** Symptom Drift Tracker
- **Theme-** HealthCare
- **PS Category-** Software
- **Team ID-**
- **Team Name: Jolly Hackers**

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IDEA TITLE

PROBLEM STATEMENT:
Symptom Drift Tracker: Long-Term Disease Evolution AI
CHALLENGE: Build a system that tracks symptom evolution over time and detects subtle health deterioration trends.

Unique Value Propositions (UVP):
1. Velocity-Based Tracking: By calculating the Drift Coefficient, the AI identifies subtle 1% weekly deteriorations that are invisible to both patients and doctors during standard 15-minute check-ups.
2. Zero-Effort Biometrics: Eliminates user burnout by silently harvesting Micro-Signals (typing cadence, gait symmetry, and voice jitter).
3. Personalized Digital Twin: Uses Self-Supervised Learning to compare a patient only to their own historical data rather than a generic population average. This ensures alerts are triggered by genuine personal decline rather than pre-existing disabilities or noise.

IDEA/SOLUTION:
A longitudinal AI engine that creates a Digital Baseline for chronic patients. It doesn't just track symptoms; it calculates the **velocity of deterioration** by correlating patient-reported outcomes (PROs) with passive biometric data.

- While traditional apps track flares, **VeloHealth** tracks drift — the subtle, 1% weekly decline in mobility, sleep quality, or cognitive load that patients adapt to and forget to report.
- We utilize **Self-Supervised Temporal Contrastive Learning**. Instead of comparing a patient to a standard human, the AI compares the patient to their own 6-month historical self to identify anomalies in their unique physiological signature.
- Correlates **biometric telemetry** with **environmental metadata** to filter out temporary "weather flares." This ensures the AI identifies true biological "drift" rather than external environmental noise.
- A **Clinician Dashboard** using "Traffic Light" triaging to prioritize high-risk patients. It only triggers alerts when the **Drift Coefficient** crosses a critical threshold, preventing "alarm fatigue" for medical providers.

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TECHNICAL APPROACH

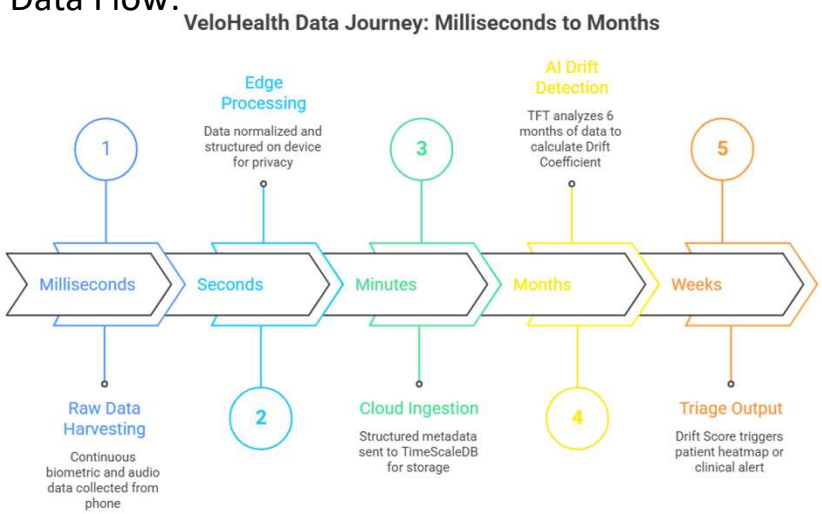
Tech Stack:

- **Backend:** Python (FastAPI), PostgreSQL (Database)
- **AI/ML:** PyTorch for **Temporal Fusion Transformers**
- **Data Integration:** APIs for EMR syncing and HealthKit/Google Fit.
- **Frontend:** Flutter (Cross-platform) for a seamless, low-friction UI.

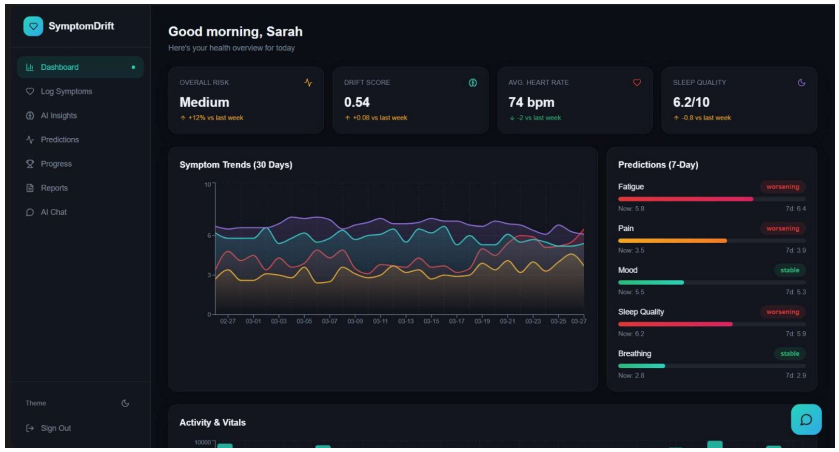
Methodology:

1. **Data Ingestion:** Passive (wearables) and Active (30-second voice/text check-ins).
2. **Feature Engineering:** Calculating the Drift Coefficient : $D = \frac{\Delta \text{Symptom Intensity}}{\Delta \text{Time}} \times \text{Variance Factor}$
3. **Visualization:** A **Heatmap of Change** rather than a standard line graph.

Data Flow:



Prototype:



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FEASIBILITY AND VIABILITY

FEASIBILITY:

- We aren’t inventing new hardware; we apply advanced time-series math to existing data streams.
- Uses industry-standard **APIs** for seamless data exchange with existing hospital Electronic Medical Records (EMR).
- Initial data processing occurs on-device, ensuring low latency, high privacy, and reduced cloud infrastructure costs.

POTENTIAL CHALLENGES AND RISKS:

- **User Fatigue:** Patients stop logging symptoms after two weeks.
- **Data Privacy :** Health data is "radioactive."
- **Signal Noise:** Differentiating a "bad day" from a "downward trend."

STRATEGIES TO OVERCOME THE CHALLENGES:

- Use **Passive Sensing** (heart rate variability, step symmetry) to minimize the need for manual input.
- Implement **Federated Learning**, where the AI model learns on the device without the raw personal data ever leaving the phone.
- Use a 14-day rolling **Gaussian filter** to smooth out daily outliers and only triggers an alert when a **consistent downward trend** is detected over two weeks.

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IMPACT AND BENEFITS

1. Support for Poor Vision: Removes visual barriers by utilizing **passive sensors** (gait/typing) and **voice-first interactions**, eliminating the need for manual text entry.

2. Family Peace of Mind: A **Care Circle Dashboard** provides loved ones with trend summaries and early-warning alerts to enable intervention before a crisis.

3. Profound Social Impact: Advances **Health Equity** for rural patients and provides objective data to validate symptoms that are often dismissed or invisible.

4. Massive Economic Benefits: Reduces hospitalizations by shifting from **expensive emergency care** to **early preventative intervention**, saving thousands in long-term healthcare costs.

5. Long-term Health Benefits: Powers **Precision Medicine** with high-resolution data, allowing doctors to personalize treatments that slow disease progression and extend quality-of-life.

6. Frictionless User Experience: Ensures **Zero-Effort Adherence** by working silently in the background, removing the "homework" of manual tracking and preventing user burnout.

Your Team Name

RESEARCH AND REFERENCES

- https://gemini.google.com
- https://pmc.ncbi.nlm.nih.gov/articles/PMC10150933/
- https://journals.mriindia.com/index.php/ijacect/article/view/464